

UNIT-IV**LINEAR ORDINARY DIFFERENTIAL EQUATIONS OF HIGHER ORDER****TUTORIAL SHEET – 1**

1. Solve the following differential equations.

a. $2y'' - 5y' - 3y = 0$

b. $(D^2 - 10D + 25)y = 0$

c. $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 7y = 0$

d. $y''' + 3y'' - 4y = 0$

e. $\frac{d^4y}{dx^4} + 2\frac{d^2y}{dx^2} + y = 0$

f. $3y''' + 5y'' + 10y' - 4y = 0$

g. $\frac{d^3u}{dt^3} + \frac{d^2u}{dt^2} - 2u = 0$

h. $y'' + 16y = 0, y(0) = 2, y'(0) = -2$

i. $y'' - 2y' + y = 0, y(0) = 0, y(1) = 3$

j. $y'' + y = 0, y'(0) = 0, y'\left(\frac{\pi}{2}\right) = 0$

k. $y'' + 2ky' + (k^2 + w^2)y = 0, y(0) = 1, y'(0) = -k$

l. $y''' - 4y' = 0, y'(0) = y(0) = y''(0) = 1$

2. The roots of the cubic auxiliary equation are $m_1 = 4, m_2 = m_3 = -5$, what is the corresponding homogeneous linear differential equation?

3. Show that the solution of the differential equation

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} - 4y = 0 \text{ is } y = e^{-2x}[c_1 \cosh \sqrt{2}x + c_2 \sinh \sqrt{2}x].$$

TUTORIAL SHEET - 2

1. Find the solution of the following differential equations:

(i) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = \cosh x$

(ii) $\frac{d^3y}{dx^3} - 2\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = (e^{3x} + 3)^2$

2. Solve the following differential equations.

a. $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = \cos^2 x + 2^x$

b. $\frac{d^2x}{dt^2} + x = \sin t \sin 2t \sin 3t$

c. $\frac{d^4y}{dx^4} - y = \cosh(x - 1) + \sin x$

d. $\frac{d^3y}{dx^3} + y = 65 \cos(2x + 1) + e^x$

e. $(2D^2 + 2D + 3)y = x^2 + 2x - 1$

f. $\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} + \frac{dy}{dx} = e^{2x} + x^3$

g. $y'' - y = x, y(0) = 0, y(1) = 0$

h. $\frac{d^2y}{dx^2} + 4y = \sin 2x, y(0) = 1, y\left(\frac{1}{2}\right) = 0$

3. Solve the following differential equations:

(i) $\frac{d^2y}{dt^2} + 4y = t^2 e^{3t}$

(ii) $\frac{d^3y}{dx^3} - 7\frac{dy}{dx} + 6y = e^{2x}(1 + x)$

(iii) $\frac{d^2y}{dx^2} + 4y = 2e^x \sin^2 x$

(iv) $\frac{d^2x}{dt^2} - 4x = t^2 \cos t$

(v) $\frac{d^2u}{dx^2} - 4u = x \sinh x$

(vi) $\frac{d^2y}{dx^2} - y = xe^x \sin x$

TUTORIAL SHEET - 3

1. Find the solution of the following differential equations:

(i) $2x \frac{d^2y}{dx^2} + 3 \frac{dy}{dx} - \frac{y}{x} = 5 - \frac{\sin(\log x)}{x}$

(ii) $x^3 \frac{d^3y}{dx^3} + 2x^2 \frac{d^2y}{dx^2} + 2y = 10 \left[x + \frac{1}{x} \right]$

(iii) $(5 + 2x)^2 \frac{d^2y}{dx^2} - 6(5 + 2x) \frac{dy}{dx} + 8y = 6x$

(iv) $x^2 \frac{d^2y}{dx^2} - 2y + x = 0, y(2) = y(3) = 0.$

2. Solve the following differential equations by the method of variation of parameters:

(i) $y'' + y = \frac{1}{1 + \sin x}$

(ii) $y'' + 2y' + y = 4e^{-x} \log_e x$

(iii) $\frac{d^2y}{dx^2} - y = e^{-2x} \cos(e^{-x})$

(iv) $xy'' - y' = (3 + x)x^2e^x$

3. In a simple harmonic motion, the amplitude of motion is 5 meters and the period is 4 seconds. Find the time required by the particle in passing between the points which are at distance of 4 meters and 2 meters from the center of the force and are on same side of it.

Also find the velocities at these points. (Equation of SHM is $\frac{d^2x}{dt^2} = -\mu^2 x$).

4. Find the current in the RLC circuit assuming zero initial current and charge and $R = 160$ Ohm, $L = 20$ Henry, $C = 2 \times 10^{-3}$ Farad and $E = 48.1 \sin(10t)$ volts.

(The differential equation for the charge Q is $L \frac{d^2Q}{dt^2} + R \frac{dQ}{dt} + \frac{Q}{C} = E$)

Objective Questions:

- The order and degree of the differential equation $y'' - (y')^{3/2} = 2$ is ____ and ____.
- The complementary function of the differential equation is $c_1 \cos 3x + c_2 \sin 3x$, then Wronskian is _____.
- $\frac{1}{D^2+1} \sin 2t =$ _____.
- Particular integral of $\frac{d^2x}{dt^2} + \frac{1}{b}(x - a) = 0$ is _____.
- If $x = e^{2t}$ is the solution of the equation $\frac{d^2x}{dt^2} - 5 \frac{dx}{dt} + kx = 0$ then value of k is _____.
- Reduce the differential equation $x^2y'' - xy' + y = \log_e x$ to a linear differential equation with constant coefficient.